# ***UNIT 7 Technology, Organization & Management(TOM)***

## ***Concept of Technology***

***Technology is the application of scientific knowledge for practical purposes. It encompasses the creation and utilization of tools, techniques, systems, and methods of organization to solve problems, improve human life, and achieve specific goals.***

### ***Key Facts of Technology:***

- Application of Knowledge
- Practical Goals
- Tools and Techniques
- Systems and Organization
- Problem Solving and Improvement
- Tangible artifacts
- The body of knowledge

## ***Approaches to Technology***

There are various ways to approach and categorize technology, depending on the focus and perspective.

### ***Key approaches:***

**1. By Application Area:**

+ Information Technology (IT):
Focuses on the management and processing of information, including computers, software, networks, and the internet.
+ Biotechnology:
Applies biological processes and organisms to create new technologies in fields like medicine, agriculture, and environmental science.
+Nanotechnology:
Deals with manipulating matter at the atomic and molecular level to create materials and devices with novel properties.
+ Medical Technology:
Encompasses tools, devices, and procedures used to diagnose, treat, and prevent diseases.
+ Agricultural Technology (Agri-tech): Focuses on improving efficiency and sustainability in farming and food production.
+ Manufacturing Technology: Deals with the design, development, and automation of

industrial production processes.
+Educational Technology (EdTech):
Explores the use of technology to enhance teaching and learning.

## 2. By Level of Innovation:

+Incremental Innovation:
Involves making small, gradual improvements to existing technologies, products, or processes.
+ Radical Innovation:
Introduces entirely new concepts, products, or processes that significantly disrupt existing markets and create new ones.
+ Disruptive Innovation:
Creates new markets and value networks by offering simpler, more convenient, or more affordable solutions to existing problems.

## 3. By its Relationship with Society:

+ Technological Determinism:
This perspective argues that technology is the primary driver of social change, shaping human behavior and societal structures.
+Social Constructivism:
This view emphasizes that technology is shaped by social, cultural, and economic factors, and its development and adoption are influenced by human values and choices.
+Socio-technical Systems: This approach recognizes the complex interplay between social and technical elements in the development and use of technology, highlighting the mutual shaping of both.

## 4. By Implementation Strategy (in Education):

+ Supplementary Use: Technology enhances traditional teaching methods (e.g., using videos alongside textbooks).
+ Complementary Use: Different technologies are used for different learning objectives based on their strengths.

+ Integrated Use: Technology is seamlessly woven into the curriculum for structured instruction.
+ Independent Use: Technology dominates the classroom with a single medium.
+ Infused Use: Technology becomes a natural and integral part of the pedagogical approach.

## 5. By Strategic Focus:

+ Technology-Push: Driven by advancements in research and development, with the aim of finding applications for new technologies.
+ Market-Pull: Driven by identified market needs and demands, with technology development focused on addressing those needs.

# Social networking:

It involves using internet-based social media platforms to connect with others. These connections can be with friends, family, colleagues, or even customers, depending on the platform's purpose. It's a way for individuals and organizations to interact, share information, ideas, and content.

Think of social networking as the digital evolution of traditional social gatherings. Instead of meeting face-to-face, people connect online, transcending geographical boundaries and time constraints.

## Key aspects of Social Networking :

### *Purpose:*

1. Social Connection: Maintaining relationships with existing friends and family and forming new ones.
2. Professional Networking: Connecting with colleagues, industry leaders, and potential employers or employees.
3. Information Sharing: Disseminating and accessing news, ideas, and personal updates.
4. Community Building: Forming groups around shared interests, hobbies, or beliefs.
5. Marketing and Branding: Businesses use social networks to build brand awareness, engage with customers, and promote products or services.
6. Entertainment: Many platforms offer entertainment through videos, games, and interactive content.
7. Learning and Education: Accessing educational resources and participating in discussions.

## Key Features ;

While features vary across platforms, common elements include:
- User Profiles: Allowing individuals and organizations to create a digital identity.
- Connections/Friends/Followers: Enabling users to link with others in the network.
- Content Sharing: Posting text updates, photos, videos, and links.
- Communication Tools: Direct messaging, commenting, and group discussions.
- News Feeds/Timelines: Displaying updates from connected users.
- Groups and Communities: Spaces for users with shared interests to connect.

## Types of Social Networking Platforms:

Social networking services come in various formats, each catering to different needs and interests. Some common categories include:

- Social Networking Sites:
Primarily focused on connecting with friends and family (e.g., Facebook).
- Professional Networking Sites: Designed for career-building and industry connections (e.g., LinkedIn).
- Microblogging Platforms: Allowing short-form text updates (e.g., X, formerly Twitter).
- Visual Sharing Platforms: Focused on sharing photos and videos (e.g., Instagram, Pinterest, YouTube, TikTok)

- Discussion Forums: Platforms for community-based discussions around specific topics (e.g., Reddit, Discord).
- Social Blogging Platforms: Combining blogging with social interaction (e.g., Tumblr, Medium).
- Messaging Apps with Social Features: Primarily for direct communication but often include broader social networking functionalities (e.g., WhatsApp, Telegram)

**In essence, social networking has become an integral part of modern life, influencing how we communicate, learn, conduct business, and interact with the world around us. It offers numerous benefits, from staying connected with loved ones to providing vast opportunities for professional growth and information access.**

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## *Use of Technology in people Management*

The use of technology has significantly transformed various aspects of people management, making HR processes more efficient, data-driven, and employee-centric. How technology is applied in different areas are as follows:

***1. Recruitment and Talent Acquisition:***

* Applicant Tracking Systems (ATS): Automate job postings, resume screening, candidate communication, and interview scheduling. AI-powered features can even analyze candidate skills and predict their fit.
* Online Job Boards and Social Media: Expand the reach of job openings and facilitate sourcing candidates with specific skill sets.
* Video Interviewing: Enables remote interviews, saving time and costs associated with

travel.
* Gamification and Assessments: Used to assess candidate skills and cultural fit in an engaging way.

## 2. Onboarding:
* Automated Onboarding Platforms: Streamline paperwork, provide new hires with essential information, and introduce them to company culture digitally.
* Learning Management Systems (LMS): Deliver online training modules for new employees to get them up to speed quickly.
* Communication and Collaboration Tools: Facilitate introductions to team members and encourage early engagement.

### 3. Human Resource Information Systems (HRIS):
* Centralized Employee Data: Store and manage all employee information, from personal details to performance records, in a single system.
* Self-Service Portals: Allow employees to access and manage their personal information, benefits, and request time off, reducing administrative burden on HR.
* Reporting and Analytics: Generate reports on various HR metrics, providing insights for data-driven decision-making.

## *4. Performance Management:*
* Performance Management Systems: Facilitate goal setting, feedback sharing, performance reviews, and development planning.
* 360-Degree Feedback Tools: Gather feedback from multiple sources to provide a comprehensive view of employee performance.
* Continuous Feedback Platforms: Enable regular check-ins and feedback exchange between managers and employees.

## *5. Learning and Development:*
* E-learning Platforms and LMS: Deliver online courses, training modules, and track employee progress.
* Virtual Reality (VR) and Augmented Reality (AR): Offer immersive training experiences for specific skills and roles.
* Personalized Learning Recommendations: AI-powered platforms can suggest training based on employee roles, skills gaps, and career aspirations.

## 6. Compensation and Benefits:
* Payroll Software: Automates payroll processing, tax deductions, and ensures accurate and timely payments.
* Benefits Administration Platforms: Allow employees to enroll in and manage their benefits online.
* Total Rewards Platforms: Provide a comprehensive view of employee compensation, benefits, and recognition programs.

### 7. Employee Engagement and Communication:
* Employee Engagement Platforms: Conduct surveys, gather feedback, and provide

tools for recognition and rewards.
* Internal Communication Platforms: Facilitate communication, collaboration, and information sharing across the organization.
* Pulse Surveys: Regularly gauge employee sentiment and identify areas for improvement.

## 8. Workforce Management:
* Time and Attendance Systems: Track employee working hours, attendance, and leave.
* Scheduling Software: Optimize employee schedules based on workload and availability.
* Workforce Analytics: Provide insights into labor costs, productivity, and absenteeism.

## *9. HR Analytics and Reporting:*
* Data Visualization Tools: Help HR professionals understand and present HR data effectively.
* Predictive Analytics: Use historical data to forecast future HR trends, such as turnover risk or talent needs.
* AI-powered Analytics: Provide deeper insights into workforce trends and support data-driven decision-making.

Benefits of Using Technology in People Management:
* Increased Efficiency and Productivity: Automating manual tasks frees up HR professionals for strategic initiatives.
* Improved Accuracy: Reduces errors associated with manual data entry and processes.
* Enhanced Employee Experience: Provides user-friendly tools and self-service options.
* Better Decision-Making: Data-driven insights enable more informed choices.
* Improved Communication and Collaboration: Facilitates seamless information sharing.
* Enhanced Compliance: Helps ensure adherence to labor laws and regulations.
* Cost Savings: Reduces administrative costs and improves resource allocation.
* Greater Scalability: Supports organizational growth and changing needs.

Challenges of Using Technology in People Management:
* Data Security and Privacy Concerns: Protecting sensitive employee information is crucial.
* Integration Issues: Ensuring new technologies work seamlessly with existing HR systems.
* Implementation Costs: The initial investment and ongoing maintenance can be significant.
* Resistance to Change: Employees and HR professionals may resist adopting new technologies.
* Lack of Technical Expertise: HR teams may need training and support to effectively use new systems.
* Over-reliance on Technology: Maintaining the "human touch" in HR processes is still important.
* Scalability Issues: Some HR tools may not be suitable for organizations of all sizes.
* Ensuring Data Accuracy and Integrity: The quality of insights depends on the accuracy

of the data.

In conclusion, technology plays a vital role in modern people management, offering numerous tools and platforms to streamline processes, enhance the employee experience, and drive strategic decision-making. However, it's crucial to implement and manage these technologies thoughtfully, considering both the benefits and potential challenges.